

Thm. Let E/F be an Extension Field, and let $\alpha, \beta \in E$ be algebraic over F .

α and β are conjugate if and only if there exists a unique isomorphism $\gamma_{\alpha, \beta}: F(\alpha) \rightarrow F(\beta)$

which fixes F and maps α to β .

Proof. Suppose that $p(x) \in F[x]$ be the irreducible polynomial having α and β as roots. Consider the Evaluation Ring Homomorphism,

$$\phi_{\alpha}: F[x] \rightarrow F(\alpha); f(x) \mapsto f(\alpha) \quad \text{and} \quad \phi_{\beta}: F[x] \rightarrow F(\beta); f(x) \mapsto f(\beta)$$

These have a same kernel, $\ker \phi_{\alpha} = \ker \phi_{\beta} = (p(x))$. This induces the Natural Iso:

$$\mathcal{Q}_{\alpha}: F[x]/(p(x)) \rightarrow F(\alpha); f(x) + (p(x)) \mapsto f(\alpha)$$

and

$$\mathcal{Q}_{\beta}: F[x]/(p(x)) \rightarrow F(\beta); f(x) + (p(x)) \mapsto f(\beta)$$

Define $\gamma_{\alpha, \beta}: F(\alpha) \rightarrow F(\beta); x \mapsto (\mathcal{Q}_{\beta} \circ \mathcal{Q}_{\alpha}^{-1})(x)$.

Since $\mathcal{Q}_{\alpha}^{-1}$ and $\mathcal{Q}_{\beta}$ fix F , so is $\gamma_{\alpha, \beta}$ and

$$\gamma_{\alpha, \beta}(\alpha) = \mathcal{Q}_{\beta}(\mathcal{Q}_{\alpha}^{-1}(\alpha)) = \mathcal{Q}_{\beta}(x + (p(x))) = \beta.$$

So proof. Conversely, observe that

$$\gamma_{\alpha, \beta}(\text{irr}(\alpha, F)(\alpha)) = \text{irr}(\alpha, F)(\beta) = 0$$

and $\gamma_{\alpha, \beta}^{-1}(\text{irr}(\beta, F)(\beta)) = \text{irr}(\beta, F)(\alpha) = 0$, so both divides each others, the monicity gives the result.